

RICHMOND COUNTY AP, NC, USA

WMO: 722243

Lat: 34.891N

Lon: 79.759W

Elev: 358

StdP: 14.51

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03738

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.7	25.9	2.6	6.4	29.5	7.7	8.3	32.1	20.9	54.9	18.6	52.0	3.2	10	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	95.4	74.8	93.0	74.0	90.9	73.4	78.5	88.1	77.6	86.5	76.8	85.3	7.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.3	139.3	81.7	75.3	134.4	80.6	74.5	130.8	79.9	42.2	88.5	41.2	86.7	40.3	85.1	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
17.2	14.8	12.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			14.5	98.4	4.4	2.3	11.4	100.1	8.8	101.4	6.3	102.7	3.1	104.3
			13.3	80.1	4.3	1.6	10.3	81.3	7.8	82.2	5.4	83.1	2.3	84.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.9	43.4	47.2	53.1	62.9	71.2	78.4	80.7	79.2	74.7	63.5	52.2	47.2
	DBStd	15.27	10.28	9.56	9.59	7.76	6.19	4.32	3.31	4.39	5.13	7.96	8.36	9.44
	HDD50	744	256	155	80	6	0	0	0	0	0	4	72	171
	HDD65	2784	670	503	384	130	22	0	0	0	3	124	392	556
	CDD50	5449	53	76	177	392	658	852	952	904	742	422	137	84
	CDD65	2011	1	4	15	66	215	402	487	439	295	77	6	4
	CDH74	18533	2	23	136	628	1849	3910	4819	4153	2394	580	30	9
	CDH80	7640	0	0	15	151	644	1768	2195	1828	910	128	1	0
Wind	WSAvg	4.8	5.0	5.6	5.9	6.1	5.2	4.6	4.2	3.6	4.3	4.3	4.3	4.6
Precipitation	PrecAvg	45.1	3.4	3.2	3.7	3.0	2.9	4.6	4.8	5.1	4.8	3.5	3.0	3.0
	PrecMax	54.1	7.2	6.6	6.1	6.1	5.7	9.5	9.1	9.5	10.7	10.6	8.1	6.0
	PrecMin	30.8	0.9	1.4	0.7	0.9	1.4	0.8	1.4	1.1	1.0	0.0	0.4	1.0
	PrecStd	5.5	1.8	1.3	1.4	1.3	1.1	2.2	2.1	2.2	2.7	2.4	1.9	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.5	77.3	82.1	87.5	93.3	98.0	97.7	99.4	93.4	88.3	77.6	74.8
		MCWB	62.6	64.2	63.0	64.6	71.0	74.2	76.1	73.9	70.7	71.6	64.7	66.1
	2%	DB	67.8	72.2	78.0	84.1	89.5	94.4	95.0	94.3	90.7	83.1	73.3	70.6
		MCWB	59.7	61.2	62.0	64.9	69.6	74.0	75.8	74.8	71.1	69.4	62.6	64.7
	5%	DB	63.9	67.7	73.8	81.4	86.6	91.7	92.6	91.4	88.5	80.4	70.2	67.0
		MCWB	57.5	58.7	60.0	63.4	69.5	73.1	75.3	74.5	71.3	68.0	61.8	63.1
	10%	DB	60.2	63.5	69.8	77.6	83.6	89.5	90.3	89.2	85.9	76.9	66.7	63.2
		MCWB	54.6	55.3	58.1	62.8	69.0	72.6	74.6	74.2	71.0	67.2	60.0	59.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.0	67.0	67.0	72.2	75.5	78.3	80.4	79.8	77.7	76.8	69.3	69.1
		MCDB	69.5	72.5	74.5	78.2	82.9	89.0	92.0	89.5	85.2	80.2	72.4	71.2
	2%	WB	62.4	64.2	65.3	69.4	73.4	77.0	79.0	78.3	76.2	74.2	66.9	66.3
		MCDB	65.3	69.3	72.9	75.8	80.9	87.4	89.3	87.3	83.2	78.3	70.1	69.2
	5%	WB	59.5	61.2	63.2	67.8	72.4	76.0	78.0	77.3	74.9	71.9	64.3	63.5
		MCDB	62.9	65.8	70.2	75.5	80.4	86.2	87.4	85.7	81.4	76.8	67.8	66.0
	10%	WB	55.2	56.9	61.1	65.9	71.1	75.1	77.0	76.3	73.7	69.3	61.6	60.1
		MCDB	58.2	61.8	67.4	73.6	79.4	84.6	85.8	84.2	79.9	74.7	66.0	62.8
Mean Daily Temperature Range	5% DB	MDBR	18.8	20.1	21.5	22.6	20.4	19.6	18.8	18.0	18.5	20.3	21.2	18.4
		MCDBR	22.9	23.7	25.5	26.6	23.9	22.9	21.3	21.1	22.4	21.8	23.2	19.4
		MCWBR	15.5	14.2	12.0	10.6	8.0	6.8	6.2	6.0	7.0	9.6	14.1	13.7
	5% WB	MCDBR	18.8	21.8	20.7	20.5	18.8	19.9	18.8	18.0	17.2	16.3	17.5	17.6
		MCWBR	15.2	16.5	12.3	10.8	7.2	6.8	6.3	5.9	6.7	8.6	13.5	14.1
Clear-Sky Solar Irradiance	taub	0.313	0.324	0.359	0.401	0.462	0.488	0.538	0.511	0.427	0.386	0.334	0.311	
	taud	2.509	2.473	2.405	2.279	2.158	2.125	2.009	2.071	2.299	2.381	2.486	2.551	
	Ebn at Noon	284	293	289	281	264	256	242	246	264	266	273	279	
	Edh at Noon	26	30	35	42	48	49	56	51	39	33	27	24	
All-Sky Solar Radiation	RadAvg	849	1054	1398	1751	1924	2029	1919	1743	1483	1218	931	737	
	RadStd	49	119	105	125	155	141	120	82	127	153	75	63	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (74 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+295	+212

Nomenclature: See separate page